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Here 4 ms is well-nigh sufficient. If we measure in 1ms units, we'll get a larger number for the time unit between two (blood) pulses, and we will then be using a larger divisor to find the rate of blood flow per minute.

3. The problem of ambient light has been explained in the text. Please read the article thoroughly and your doubts will be cleared.

Speedometer-cum-Odometer

I have constructed the 'Microcontroller-based Speedometer-cum-Odometer' project published in Nov. 2008 issue, as per the instructions given in the text. I have downloaded the 'hex' file but nothing was displayed on the LCD screen. Also, the Assembly program is not clear. Please help.

Anil Patil
 Ulhasnagar

□ Regarding this project, I have the following doubts:

1. How to make the LCD module illuminate?
2. What is the name of the box that encloses the LCD module and the circuit board, and how to fix it in the handle-bar?
3. Do I need to provide water-resistive coating for the reed switch?
4. What changes are to be made in the program and the circuit, so that it can be used in my Honda Unicorn bike?

Roshan George V.
 Cochin

The author Arun Kumar Vadla replies to Anil Patil:

If you are testing the project on a simulator software, nothing will display on the LCD. You have to test the project in real hardware after programming the microcontroller, first with 'init.eeprom.hex' program and then with 'speedo.hex' program. The Assembly code is well commented. You'll find explanation of each line of the code. I have been using this speedometer since one-and-a-half years and it is working well without any problem.

Reply to George Roshan:

1. Backlight connections are usually made at pins 15 and 16 of the LCD module. The anode (pin 15) should be connected to +5V DC through a series

resistor. Connect the cathode (pin 16) to GND.

2. There is no particular name for the enclosure, but I have used the enclosure of an electronic choke and made a slot according to LCD dimensions.

3. Since reed switch is encapsulated in glass, it does not require any special coating.

4. In the article, I have given the formula for modifying the program for any wheel dimensions. First, calculate the radius of the wheel and then use the formula given in the text to get the constant. I have illustrated these steps in the article, so please read carefully.

Mobile Bug

Regarding the 'Mobile Bug' project published in January 2008 issue, I have the following queries:

1. It is mentioned in the article that the circuit can sense the presence of an activated mobile phone. Does it mean a switched-on mobile phone without call/SMS/data transmission?
2. Can we use an FM radio's telescopic antenna in place of the 12.7cm (5-inch) antenna shown in the circuit diagram?

Prakash
 New Delhi

The author D. Mohan Kumar replies:

1. The mentioned circuit senses an activated mobile phone. That is, switch on the circuit and it will detect the signals in response to a call, SMS or video recording in the mobile phone.

2. A small telescopic antenna can be used. The antenna is not necessary if the 0.22μF capacitor is protruding from the case.

Staircase Light

In the 'Staircase Light With Auto Switch Off' circuit published in January issue, the schematic diagram is wrong. Pin 2 of IC 555 appears at two locations.

Ramasamy
 Mysore

EFY: Thanks for pointing out the printing error in the circuit diagram. In the circuit tested, pins 6 and 7 of 555 timer are shorted at the junction of resistor R7 and capacitor C4. Kindly read wrongly-printed pin 2 as pin 7.